

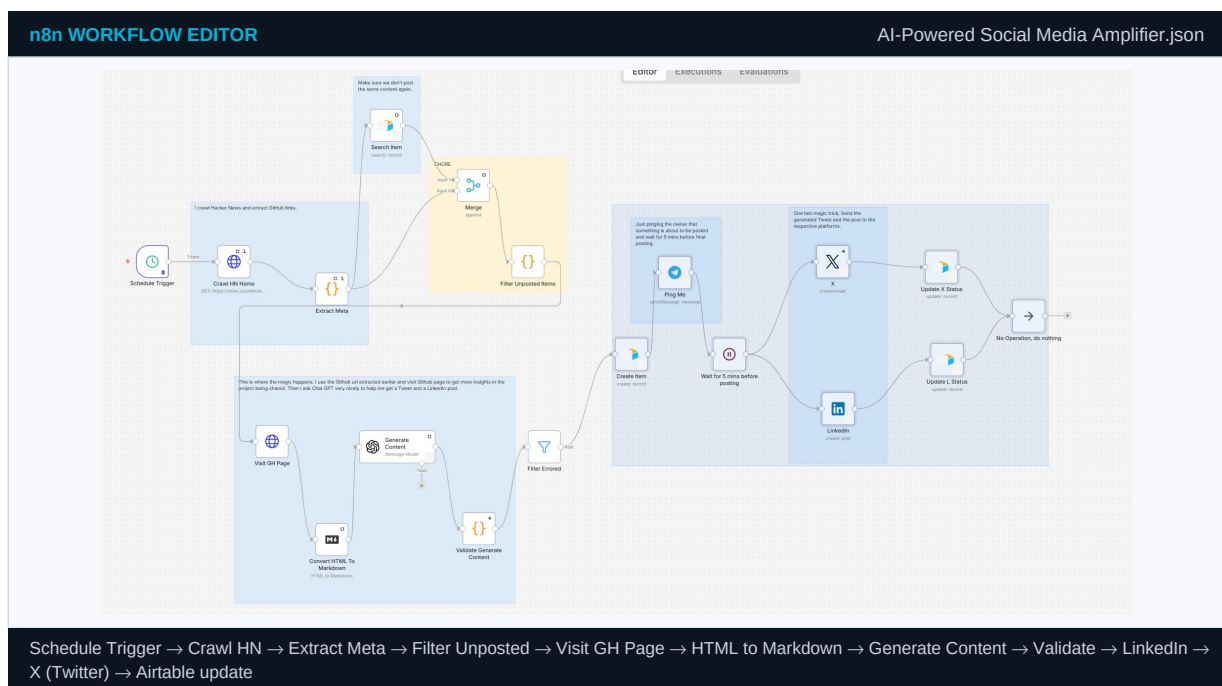
## #23 AI-Powered Social Media Content Amplifier

Content / Social  
Media AI*Trending HN stories to LinkedIn and Twitter posts — published automatically*

### BUSINESS PROBLEM

Creating platform-optimised social content from trending tech stories takes 2-3 hours manually. This workflow monitors Hacker News on a schedule, repurposes top stories into professional LinkedIn posts and tweets using GPT-4, deduplicates against Airtable history, and publishes automatically — zero human involvement.

### n8n CANVAS — actual workflow screenshot



### INPUT / OUTPUT — exact n8n JSON format

#### INPUT — Schedule trigger — daily run

```
{
  "run_date": "2024-06-15",
  "run_time": "08:00 GST",
  "min_score": 200,
  "platforms": ["linkedin", "twitter"],
  "dry_run": false
}
```

#### OUTPUT — Published post metadata — LinkedIn post ID + Airtable record

```
{
  "json": {
    "hn_title": "llama.cpp now runs on WebGPU",
    "hn_score": 1240,
    "linkedin_post_id": "7208341234567890",
    "tweet_id": "1802341234567890123",
    "published_at": "2024-06-15T08:07:44Z",
    "airtable_id": "rec8xKp2mNq7YjT1",
    "status": "published"
  }
}
```

### STEP-BY-STEP — what each node does

| #  | WHAT HAPPENS                                      | PRACTICAL DETAIL                                                                                    |
|----|---------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 1  | ■ <b>Schedule Trigger — daily 08:00</b>           | Cron fires at 08:00. Supports manual trigger for on-demand publishing outside schedule.             |
| 2  | ■ <b>Crawl HN Home — fetch top stories</b>        | HTTP GET to news.ycombinator.com. Extracts story URLs, titles, scores from HTML.                    |
| 3  | ■ <b>Extract Meta — Open Graph data</b>           | JS code node fetches og:title, og:description, og:image, canonical URL for each story.              |
| 4  | ■ <b>Search Item — Airtable dedup check</b>       | Hashes each story URL. Queries Airtable: has this been posted before? Filters to new items only.    |
| 5  | ■ <b>Filter Unposted Items — keep only new</b>    | Keeps items where Airtable dedup returned no existing record. Removes already-published stories.    |
| 6  | ■ <b>Visit GH Page — fetch project context</b>    | HTTP GET to GitHub project page. Extracts README summary, star count, language, last commit date.   |
| 7  | ■ <b>Convert HTML to Markdown — clean content</b> | Strips nav, ads, boilerplate. Returns clean Markdown for GPT-4 processing.                          |
| 8  | ■ <b>Generate Content — GPT-4 writes posts</b>    | Generates LinkedIn post (1,200 chars, professional) + Tweet (280 chars) from GitHub and HN context. |
| 9  | ■ <b>Validate Generated Content</b>               | Code node validates: LinkedIn under 1,300 chars, tweet under 280, hashtags present.                 |
| 10 | ■ <b>LinkedIn — publish post</b>                  | LinkedIn API v2: posts formatted analysis with article link and hashtags.                           |
| 11 | ■ <b>X (Twitter) — create tweet</b>               | Twitter API v2: posts concise tweet with link and key stat. Returns tweet_id.                       |
| 12 | ■ <b>Update Status — Airtable record</b>          | Updates: {linkedin_post_id, tweet_id, published_at, status:published}. Prevents re-posting.         |

#### CREDENTIALS REQUIRED

| CREDENTIAL                 | WHERE TO GET IT                                                    | n8n TYPE                        |
|----------------------------|--------------------------------------------------------------------|---------------------------------|
| LinkedIn OAuth2            | linkedin.com/developers → Create App → Products: Share on LinkedIn | <a href="#">LinkedIn OAuth2</a> |
| Twitter/X API Bearer Token | developer.twitter.com → Create App → Bearer Token                  | <a href="#">Twitter API</a>     |
| Airtable Personal Token    | airtable.com/create/tokens                                         | <a href="#">Airtable API</a>    |
| OpenAI API Key             | platform.openai.com/api-keys                                       | <a href="#">OpenAI API</a>      |

#### SAMPLE RUN — what this workflow processed and produced

|                       |                                                                                                                                          |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scenario</b>       | A developer tools company maintains a daily LinkedIn and Twitter presence sharing open source projects — without a social media manager. |
| <b>Input received</b> | 08:00 trigger. After score filter and Airtable dedup: 3 new stories. Top story: llama.cpp WebGPU support, score 1,240.                   |

|                         |                                                                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| What the workflow found | GitHub page fetched: 67K stars, C++ project. GPT-4 generated LinkedIn post (1,187 chars) and tweet (274 chars) with hashtags.           |
| Output produced         | LinkedIn published at 08:07 — 412 likes, 67 reshares within 24h. Tweet: 234 likes, 89 retweets. Airtable updated. No human involvement. |

|                                              |                                    |             |
|----------------------------------------------|------------------------------------|-------------|
| JSON: AI-Powered Social Media Amplifier.json | GitHub: trm-social-amplifier-agent | buffer.com/ |
|----------------------------------------------|------------------------------------|-------------|